Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation

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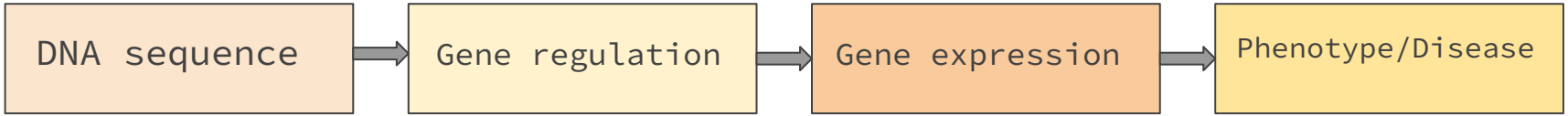
Why predict gene expression from DNA?

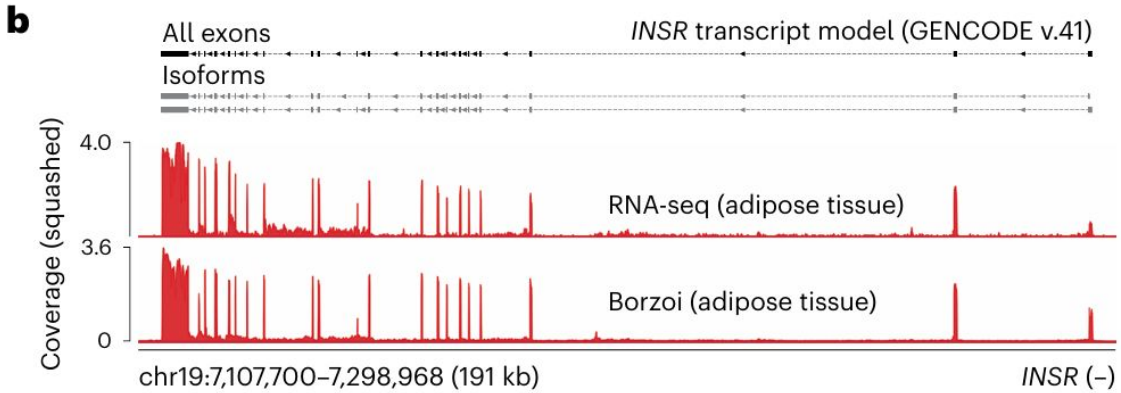


How does DNA sequence determine gene expression? Most disease-associated variants lie in non-coding regions, meaning they don’t change proteins directly, but instead affect gene regulation.

Mutations can change gene expression. Most mutations are in non-coding regulatory regions. These gene expression changes are needed to interpret GWAS signals and disease variants. The goal is to learn the mapping from DNA sequence to gene function; if we can predict gene expression directly from the DNA, we can understand which mutations are actually functional.

What is RNA-seq coverage?

RNA sequencing, or RNA-seq, measures how much RNA is produced from different parts of the genome. Instead of giving just a single number per gene, it provides a coverage profile — meaning how many sequencing reads map to each position. So what we get is a curve over the genome, where peaks correspond to expressed regions, typically exons.



RNA-seq coverage prediction for the held-out test gene INSR (GTEx ‘adipose tissue’), obtained by averaging the predictions of four model replicates. The ‘squashed’ scale refers to the transformed scale applied to the training data (Methods).

Why is RNA-seq prediction difficult?

Existing models:

Some of the existing models predict:

- TF binding (ChIP-seq)
- Accessibility (ATAC-seq)

They work on local regions (500 - 2000 bp)

RNA-seq:

Depends on very long regions (100kb - 500kb). The signal is not local.

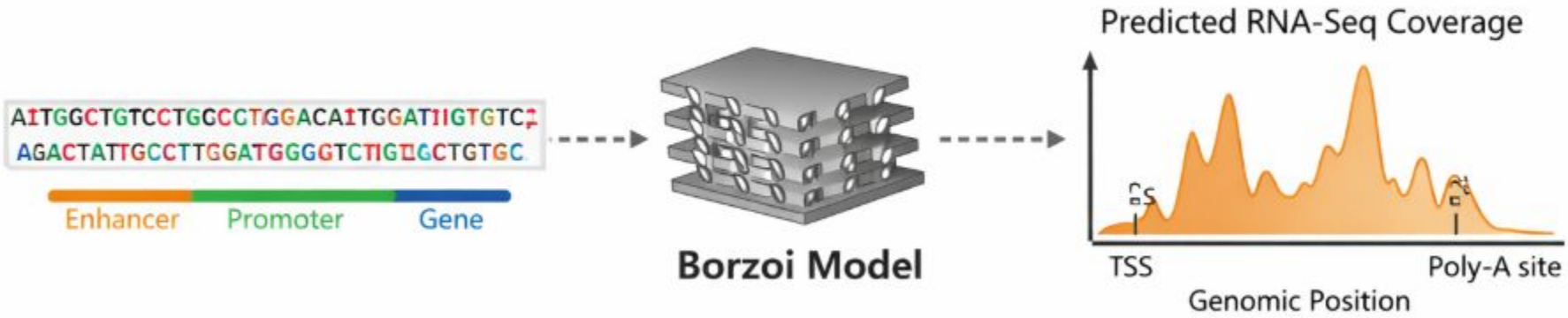
It integrates:

- Transcription
- Splicing
- Polyadenylation
- RNA stability

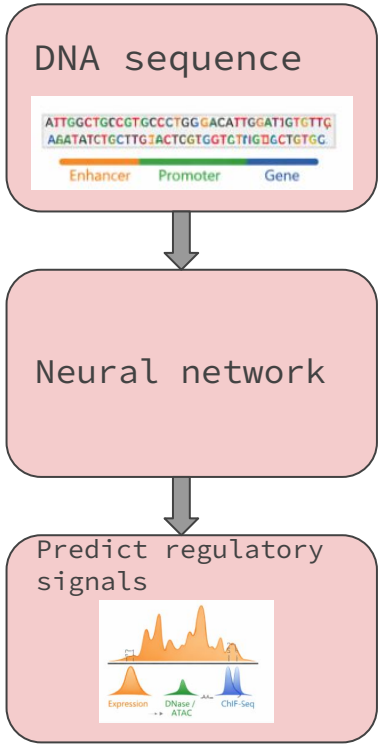
Contribution of the paper

Our problem is that the RNA-seq is complex and long-range.

This work proposes the following solution: The Borzoi model takes the long DNA sequence as input, learns expression, splicing, and polyadenylation, and outputs the RNA-seq coverage. There is no need for gene annotations.



What did the previous models do?



One of the most important prior models is the Enformer model.

It uses self-attention, which enables the model to capture long-range interactions across the genome.

It predicts:

- CAGE (expression at TSS)
- DNase / ATAC (accessibility)
- ChIP-seq (TF binding)

However, it cannot directly model RNA-seq coverage across entire transcripts.

From Enformer to Borzoi

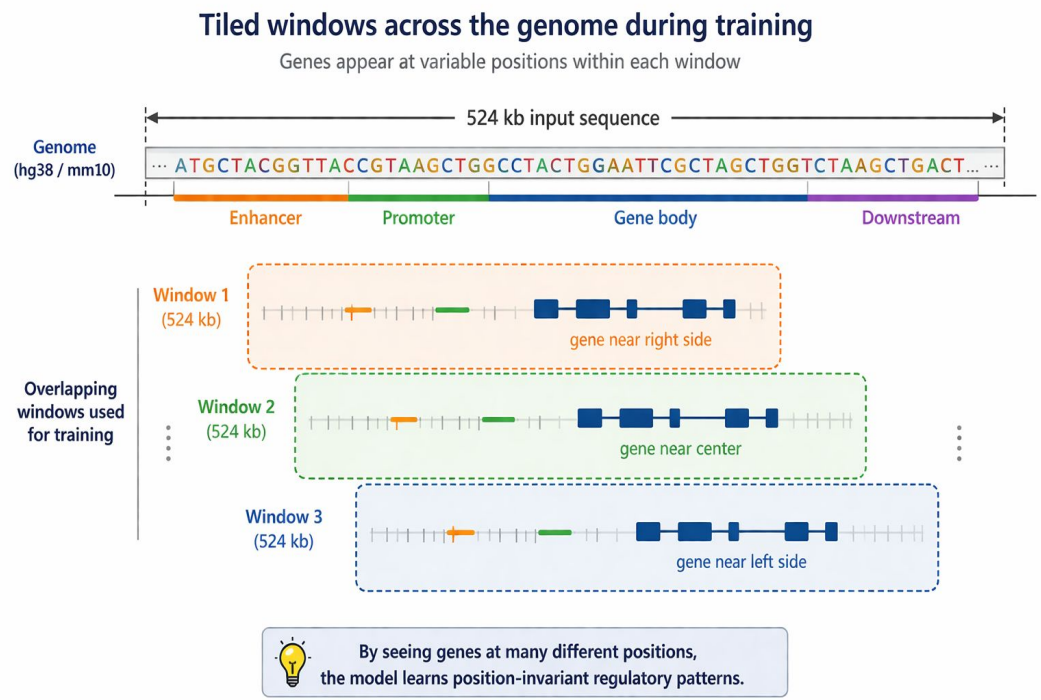
Enformer	Borzoi
Predicts regulatory signals	Predicts RNA-seq coverage
Focus on TSS-level signals	Full transcript coverage
Attention-based	Attention + U-Net

Key Biological Concepts

- **eQTLs** — expression quantitative trait loci — are genetic variants that affect how much of a gene gets transcribed. These are the primary benchmark for evaluating how well a model can predict variant effects on expression.
- **sQTLs** — splicing QTLs — affect how a gene is spliced, changing which exons are included. Predicting these requires the model to understand the full exon-intron structure of a gene.
- **paQTLs** — polyadenylation QTLs — affect where the transcript ends, by altering which poly-A site is used. This determines 3' UTR length and can affect mRNA stability and translation.

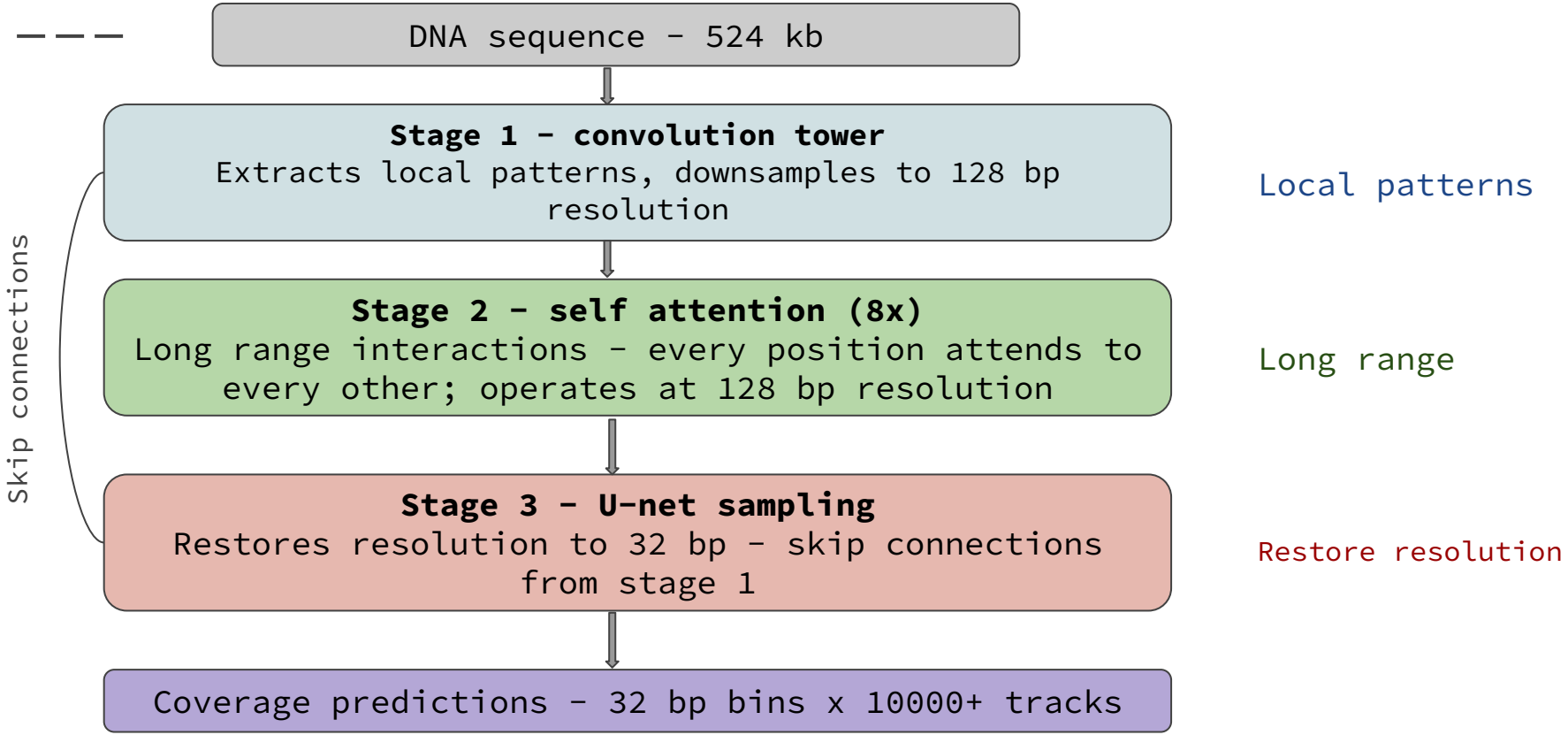
Borzoi is one of the first models to handle all three of these simultaneously from raw sequence.

The Input- what does Borzoi actually see?

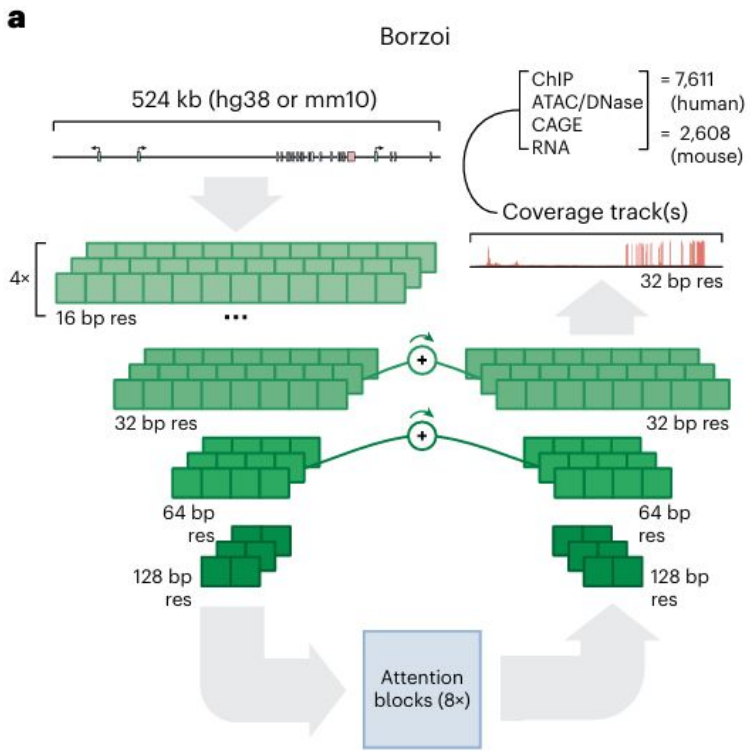


- One-hot encoded DNA sequence (A/C/G/T → 4-channel vector)
- Tiled windows across the genome during training
- Genes appear at variable positions within each window

The architecture - a three stage pipeline

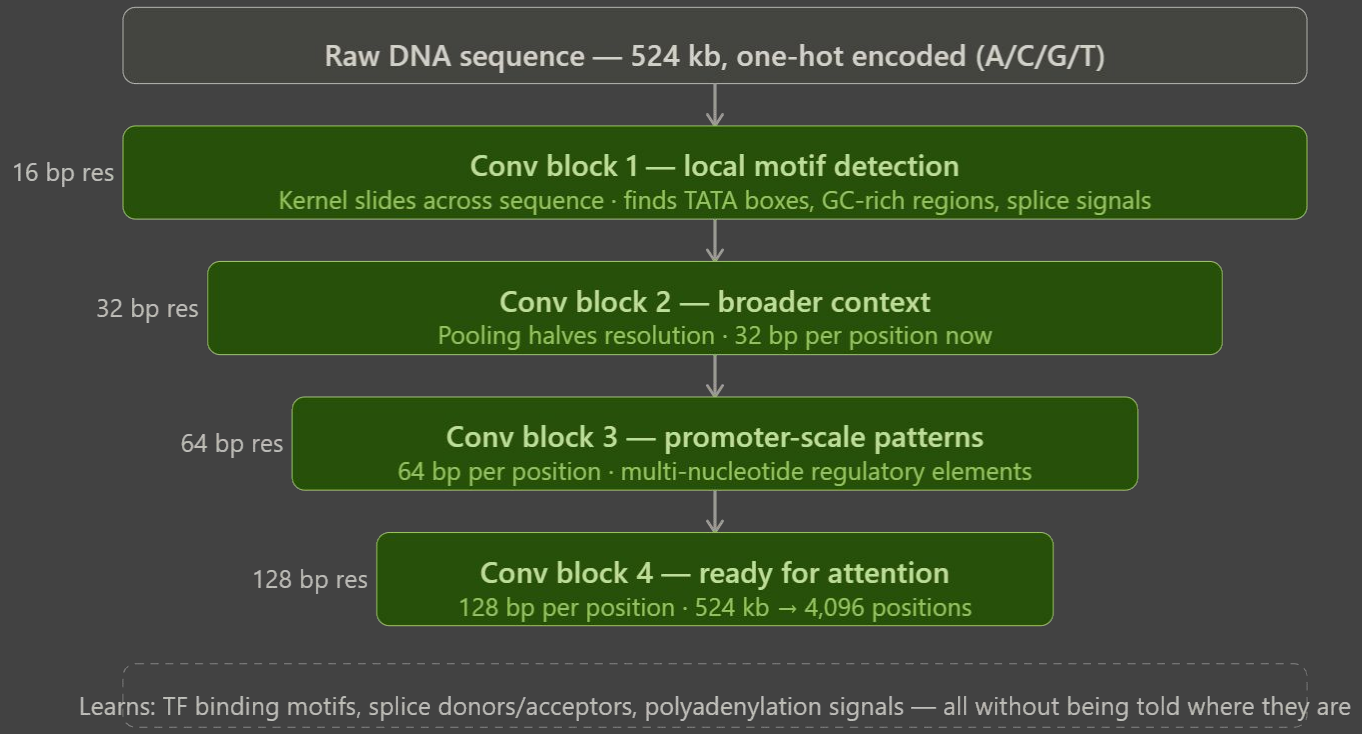


The architecture - a three stage pipeline



Borzoi: a neural network for predicting RNA-seq coverage from sequence. a, The Borzoi neural network architecture consists of a number of convolution and downsampling layers followed by a stack of self-attention layers with relative positional encodings operating at 128 bp resolution, similar to the Enformer architecture. The output is then repeatedly upsampled and put through additional convolution layers with matched U-net connections to predict at 32 bp resolution. Connections with '+' symbols represent a combination of the outputs of a previous layer with the inputs of a new layer through residual convolution.

Stage 1 - a convolution tower



Stage 2 - self attention

4,096 positions at 128 bp resolution (= 524 kb)

Line thickness = attention weight (how strongly one position influences another)

8 transformer blocks — each refines the representation

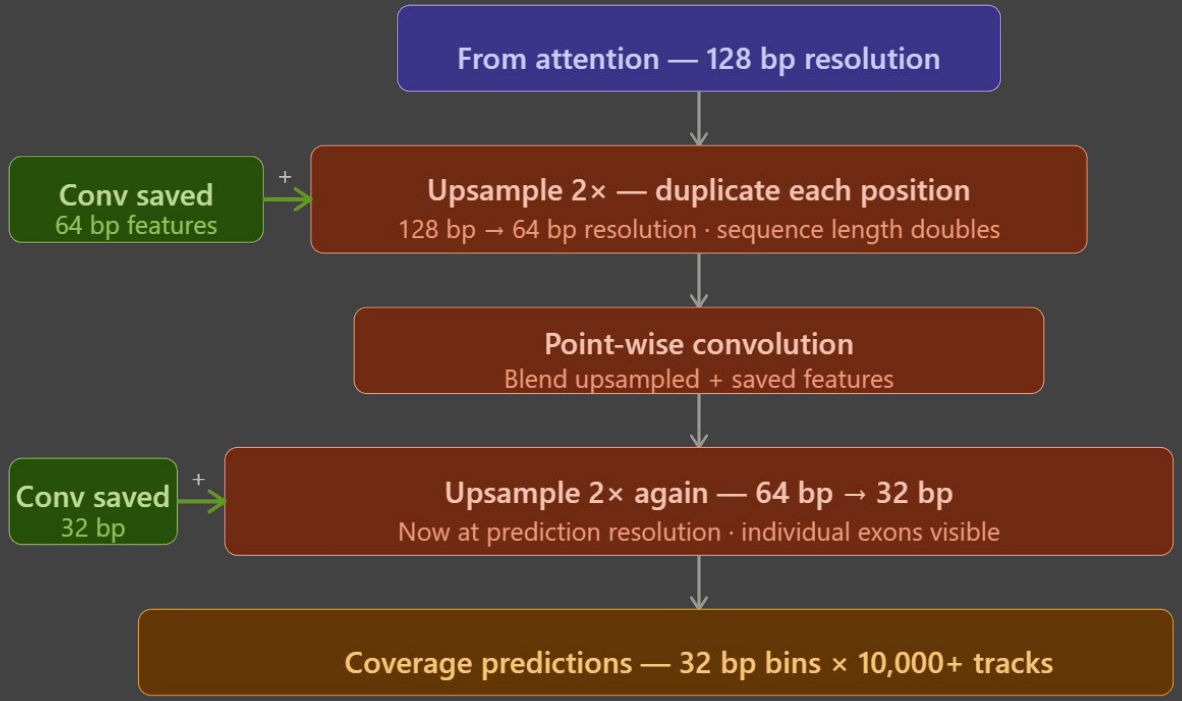
Block 1 — initial long-range context

Block 2 — refine relationships

Block 3 · · · Block 8 — hierarchical regulatory reasoning

Why this matters for RNA-seq: a read at the 3' end of a gene depends on an enhancer 200 kb upstream. Convolution cannot bridge that gap. Attention can.

Stage 3 - U-Net sampling



Without skip connections the model loses exon boundary sharpness — splice sites blur into background noise

What does Borzoi predict?

What is predicted:

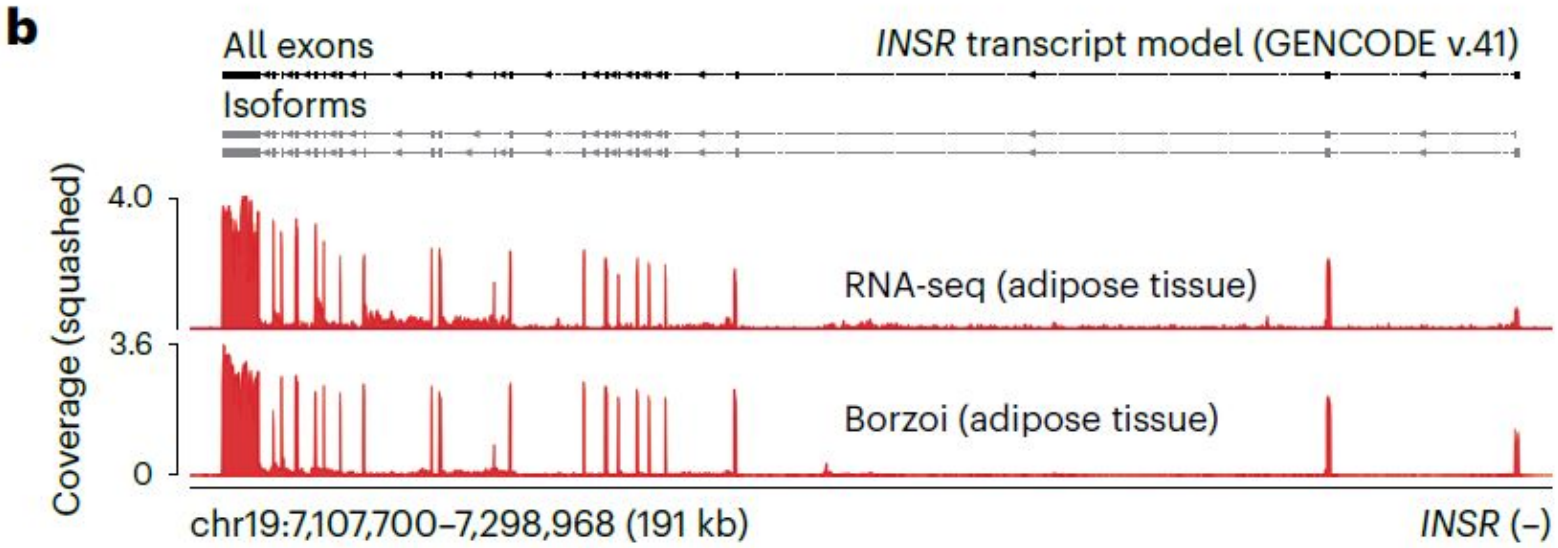
- Multi-task prediction of genomic signals
- RNA-seq + regulatory assays

Scale of the data:

- 866 human + 279 mouse RNA-seq datasets
- ~7,600+ human tracks total

NOTE: RNA-seq normalized using power transform (y^(3/4))

Results- predicting RNA-seq coverage from sequence.

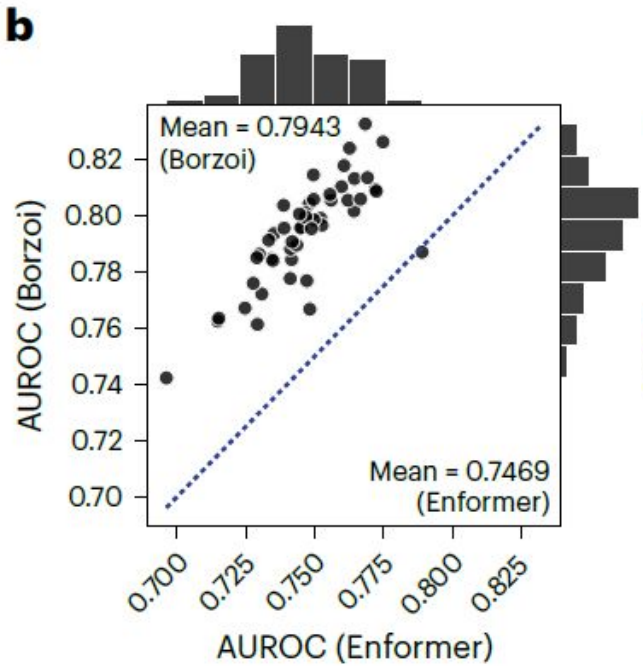


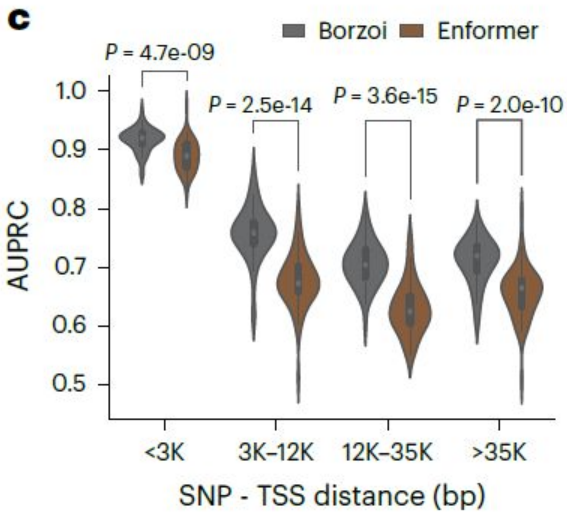
RNA-seq coverage prediction for the held-out test gene INSR (GTEx ‘adipose tissue’), obtained by averaging the predictions of four model replicates.

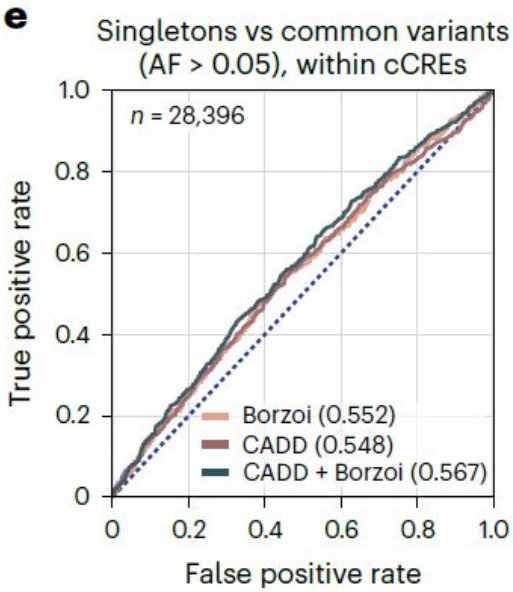
Insights

- **Direct Translation:** Borzoi successfully translates raw DNA sequences directly into high-resolution RNA-seq coverage patterns without needing external gene annotations.
- **High Predictive Accuracy:** The model's predictions strongly correlate with real experimental data, both at the fine-grained level (specific genomic bins) and the macro level (total gene expression).
- **Captures True Biological Context:** The model doesn't just memorize baseline averages; it accurately predicts tissue-specific variations
- **Scales to Complex Systems:** Despite the difficulty of long-range modeling, Borzoi handles massive mammalian genes with complex exon/intron structures over 500+ kilobase windows.

Results-predictions of variant effects align with eQTL results and negative selection



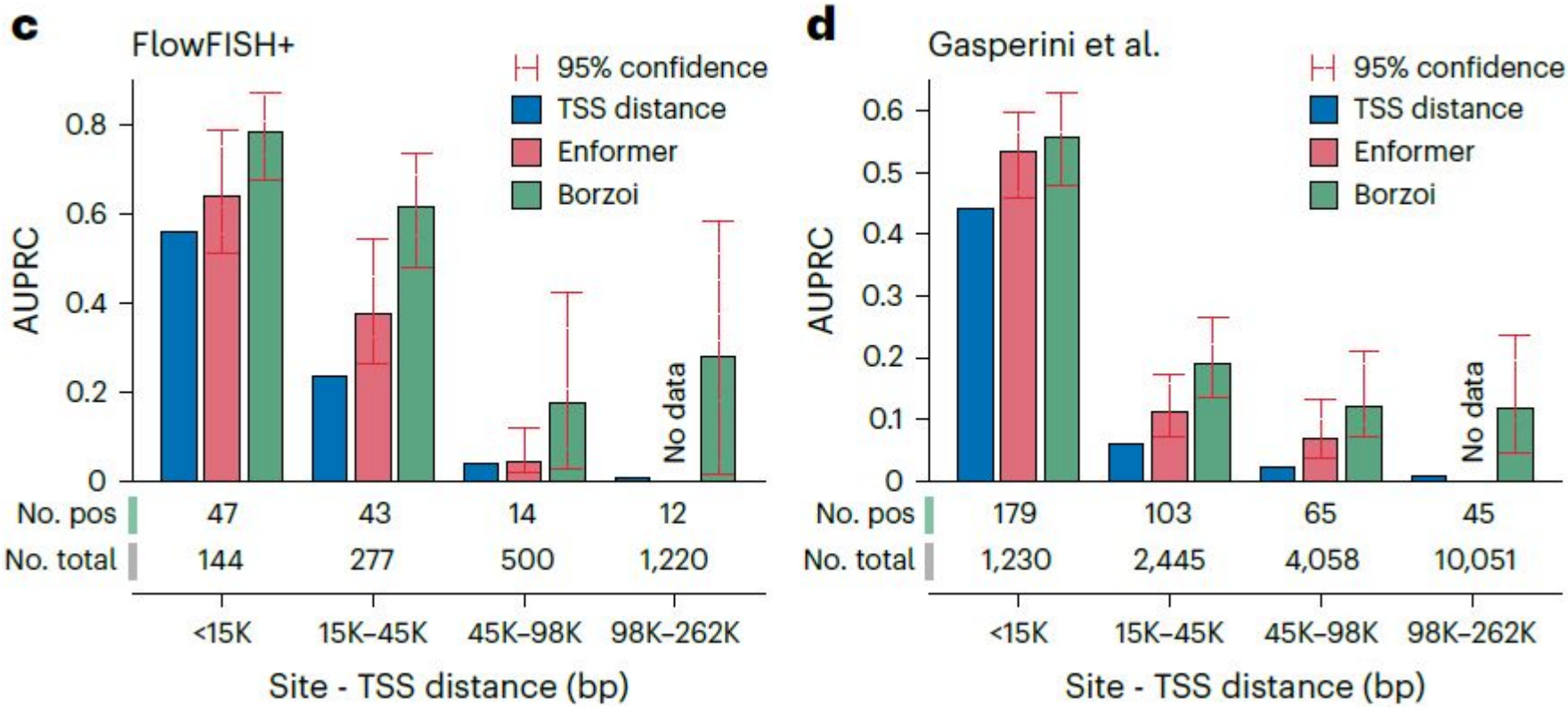




Insights

- Borzoi systematically outperforms previous leading models (like Enformer) at classifying known functional mutations (eQTLs) across almost all tissue types.
- **Detects Long-Range Errors:** While older models are okay at finding errors right next to a gene's start site, Borzoi is vastly superior at detecting functional mutations located very far away.
- **Predicts Impact Severity:** The model doesn't just flag a mutation; it more accurately predicts the *magnitude* of the effect that the mutation will have on overall gene expression.
- **Identifies Dangerous Anomalies:** Borzoi effectively acts as an anomaly detector, reliably distinguishing rare, potentially pathogenic "singleton" mutations from common, harmless human variations.

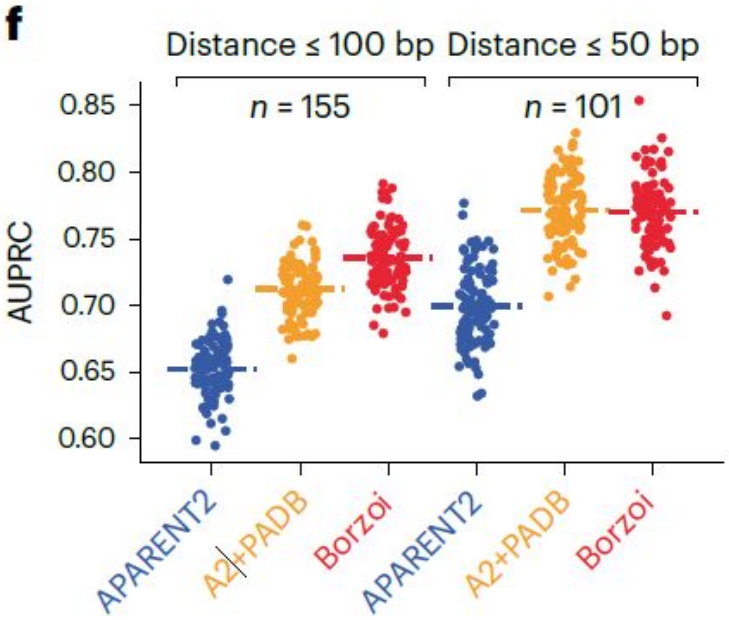
Results-Predicting the impact of context and distal regulatory elements on gene expression.

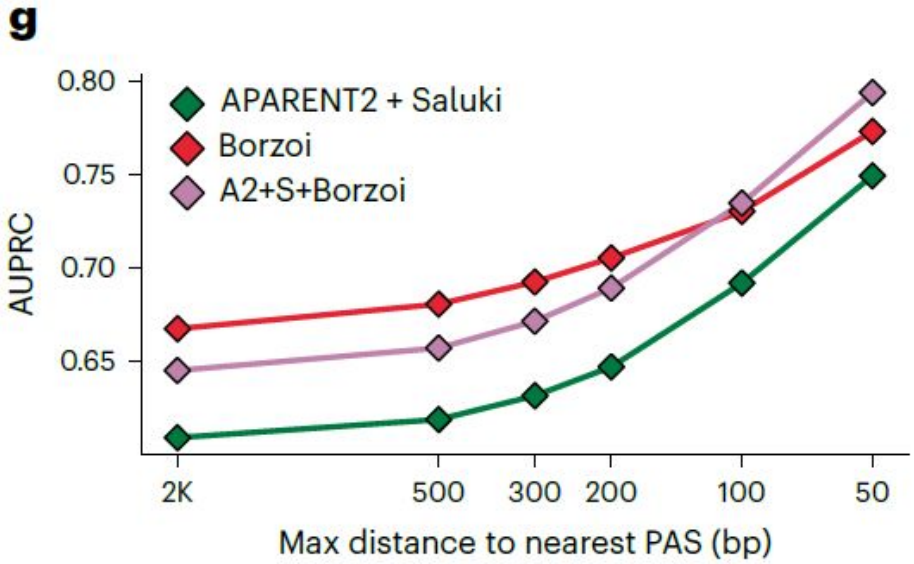


Insights

- **Maps Distant Configurations**: Borzoi successfully identifies which distant regulatory elements (CREs) control which specific target genes, mapping out the system's hidden architecture.
- **"White-Box" Interpretability:** By extracting internal mathematical gradients (saliency), the model reveals exactly *how* a change at a distant switch mathematically impacts the target gene's output.
- **Excels at Long-Range Dependencies:** The model remains highly accurate even when the control switch is located hundreds of thousands of base pairs away from the gene it regulates.
- **Significant Performance Leap:** Across multiple independent datasets, Borzoi demonstrates a massive improvement over prior architectures in predicting these complex, long-range regulatory relationships.

Results - Predicting APA and 3′ polyadenylation QTLs





Insights

- **Generalist Beats Specialists:** Despite being a unified, multi-task model, Borzoi performs competitively with—and often outperforms—narrow, highly specialized state-of-the-art models (like APARENT2) built solely to predict RNA termination.
- **Masters Long-Range Context:** Specialized tools fail when a mutation occurs far away from the termination signal. Because Borzoi processes massive 524kb windows of sequence, it successfully captures distant contextual rules that dictate where transcripts end.
- **Accurately Predicts paQTLS:** The model reliably acts as a classifier for paQTLs, successfully identifying which genetic variants will disrupt normal transcript termination and which are harmless.
- **Synergistic Ensembling:** While Borzoi is highly capable on its own, the authors demonstrated that combining Borzoi's long-range contextual awareness with the localized precision of specialized tools (creating an Ensemble) yields the highest overall predictive accuracy.

Splice Site Prediction Implementation

Data details and preprocessing

- **Dataset**: Human reference genome Chromosome 22 (GRCh38/hg38) fetched directly via the **Ensembl REST API**.
- **Encoding**: Each **400 bp** window is **one-hot encoded** into a 4-channel vector ($A, C, G, T \rightarrow$ 4-channel vector).
- **Coordinate Source**: Exon coordinates and sequence data for protein-coding genes were obtained using Ensembl's `overlap/region` and `lookup/id` API endpoints.
- **Label/Target Generation**:
  - **Positive Samples**: Genomic windows centered on annotated **splice donor** (exon end) and **acceptor** (exon start) sites.
  - **Negative (Decoy) Samples**: Background sequences from Chromosome 22 containing **"GT"** or **"AG"** motifs that are not located within 5 bp of any annotated splice site.
- **Data Splits**: The dataset (containing 62,220 total samples) was divided into **80% training** and **20% testing** subsets using a randomized split.

Model Architecture

- **Architecture Type:** 3-layer Convolutional Neural Network (CNN).
- **Input Layer:** Processes 200bp/400bp DNA sequences, one-hot encoded into a 4-channel matrix (A, C, G, T).
- **Layer Structure:** * Three consecutive **1D Convolutional layers** for feature extraction.
    - **Batch Normalization** after each convolution to stabilize training and improve convergence.
    - **ReLU Activation** functions to introduce non-linearity.
    - **Max Pooling** to downsample and identify the most prominent features (motifs).
- **Final Layer:** A Fully Connected (Linear) layer with a Sigmoid activation for binary classification (Splice Site vs. Decoy).
- **Parameter Count:** ~103,489 parameters (a "lightweight" and efficient model compared to Borzoi).

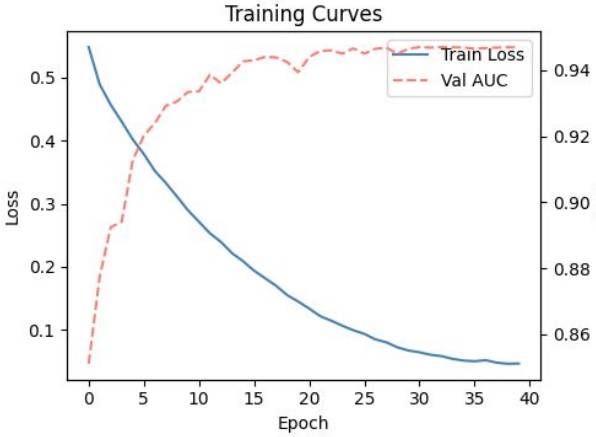
Training Strategy and Metrics

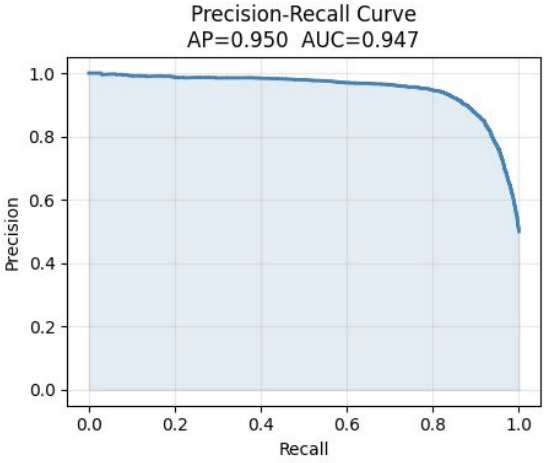
- **Optimization:** Trained using the **Adam Optimizer** with **Binary Cross-Entropy (BCE) Loss**, standard for binary classification tasks.
- **Training Scale:** ~50,000 sequences used for training, with a 20% hold-out set for testing.
- **Hardware:** Trained on a **Google Colab T4 GPU** (taking advantage of parallel computing for fast convolution operations).

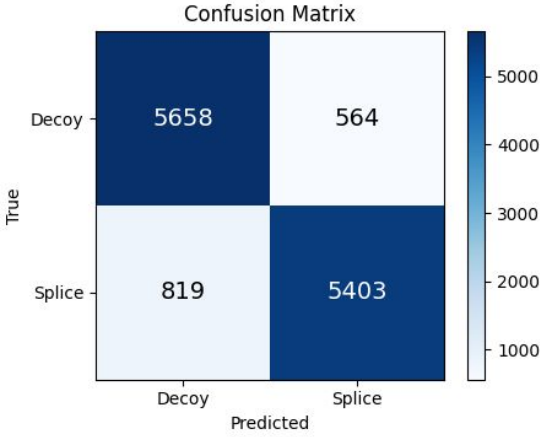
Why this architecture

- **Mimicking Borzoi’s "Stage 1":** Borzoi’s architecture begins with a "Convolution Tower" designed to extract local patterns. Our 3-layer CNN acts as a **miniature version of Stage 1**, proving that a local convolutional approach is the correct "front-end" for biological sequence interpretation.
- **Addressing Compute Constraints:** The full Borzoi model requires enterprise-grade hardware (A100 GPUs) and massive memory. By building a 3-layer CNN, we created a **hardware-agnostic validation** that still achieves high performance on consumer-grade Colab environments.
- **Specialization over Generalization:** While Borzoi is a "generalist" (predicting everything), it can be a black box. We used this specific architecture to **isolate the splicing problem**, allowing us to verify that a model can learn specific biological logic gates (splice sites) without needing the full complexity of a Transformer.
- **Feature Locality:** Since splicing signals (like the GT/AG motifs) are concentrated in a small local window, a CNN is the most efficient mathematical tool for the job. It utilizes "parameter sharing," meaning the same filters scan the whole sequence to find relevant motifs, regardless of their exact position.

Results







Test accuracy = 0.889

Motif visulaization

Observations

Decoding the "Black Box": The motif visualizations (generated via In-silico Mutagenesis) act as a visual debugger, mapping the learned weights of our CNN to reveal exactly which sequence patterns drive the model's predictions.

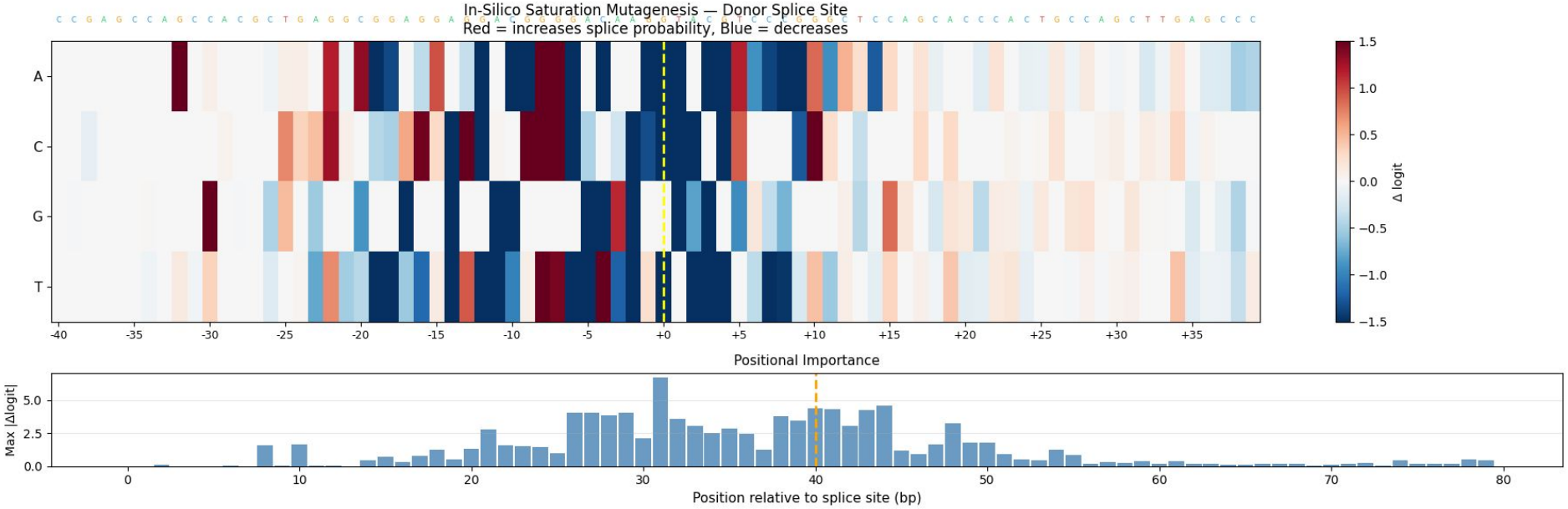
Identifying Biological "Logic Gates": The saliency maps show a massive spike in importance exactly at the **"GT"** (donor) and **"AG"** (acceptor) positions, confirming the network successfully learned the primary biological logic gates for splicing without hardcoded rules.

Capturing Broader Context: Beyond the core 2-letter motifs, the visualizations highlight the importance of surrounding "flanking" sequences, proving the model evaluates complex genomic context rather than just memorizing simple string matches.

Foundation for Variant Effect Prediction (VEP): By mathematically scoring the sensitivity of every single position, these maps create the essential "lookup table" needed to predict whether a specific DNA mutation will act as a harmless typo or cause a functional "system crash."

Mirroring Borzoi's Discoveries: This local analysis mirrors the Borzoi paper’s methodology (Figure 3), demonstrating how deep learning models can autonomously discover and validate fundamental cis-regulatory code directly from raw sequences.

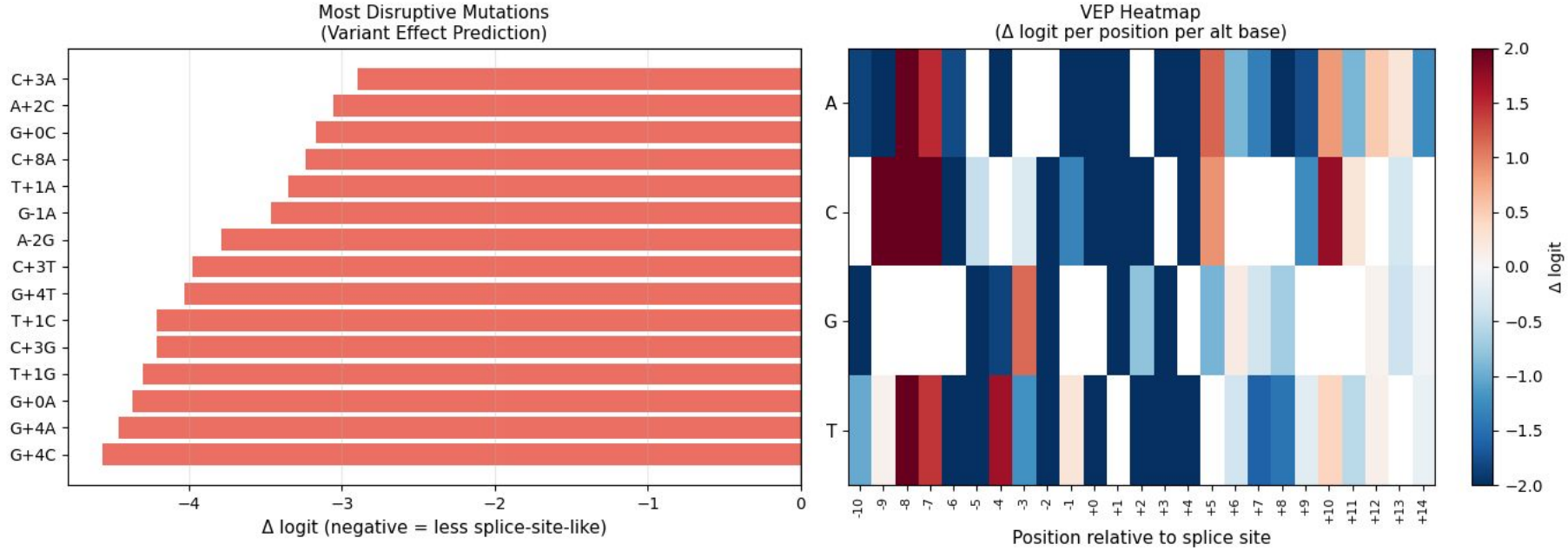
ISM Plot


In-Silico Saturation Mutagenesis — Donor Splice Site
Red = increases splice probability, Blue = decreases
A
C
G
T
-40
-35
-30
-25
-20
-15
-10
-5
+0
+5
+10
+15
+20
+25
+30
+35
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Δ logit
Positional Importance
Max |Δlogit|
5.0
2.5
0.0
0
10
20
30
40
50
60
70
80
Position relative to splice site (bp)

Observations

- **Reading the plot:**
  X-axis = sequence position; Y-axis = importance score (Δ).
  Large negative values → mutation strongly reduces model confidence.
- **Key Insight – "Logic Gate":**
  Sharp spike at central **GT/AG** shows the model correctly identifies core splice signals without explicit rules.
- **Beyond Simple Patterns:**
  Secondary spikes in flanking regions (e.g., polypyrimidine tract) indicate the model uses broader sequence context—not just 2-base motifs.
- **Connection to Borzoi:**
  Mirrors the ISM approach in Borzoi (Fig. 3), which discovers long-range regulatory elements from raw DNA.
- **Why It Matters:**
  Forms the basis of Variant Effect Prediction (VEP): quantifies how mutations impact RNA splicing.

Variant effect Prediction



Observations

- **Reading the plot:**
  Log2 Fold Change (Δ score); 0 = neutral, large negative = disruptive mutation.
- **Silent vs. Critical Mutations:**
  Model distinguishes neutral variants from pathogenic ones (hits at **GT/AG** sites).
- **Positional Sensitivity:**
  Same mutation can be high-impact at splice site, negligible just ~20bp away.
- **Connection to Borzoi:**
  Aligns with zero-shot variant prediction in Borzoi (Fig. 5–6).
- **Why It Matters:**
  Δ score quantifies functional loss → predicts impact of genetic variants on RNA processing.

THANK YOU

Github link - https://github.com/soum-stacks/Splice_site_prediction